Control work 10 — three-dimensional shapes

Counts, nets, surface areas, volumes and views, in twenty marks.

LESSON 189 | 1 × 40 MIN | MATEMATIKA 6, NAZORAT ISHI 10 | S7 8 REVIEW

LESSON CLOCK

2' | 28' | 8' | 2'

Instructions	2 min
The paper	28 min
Answers and diagnosis	8 min
What comes next	2 min

LEARNING OBJECTIVES

- Give the faces, edges and vertices of a named solid.
- Find a surface area and a volume for a cuboid, a prism and a cylinder.
- Describe the net of a cylinder with its measurements.
- Name each lost mark by its type and rewrite the whole solution.

TEXTBOOK

National	pp. 498–541
Cambridge	Stage 7, pp. 84–96
Workbook	Control paper 10

01 Explanation

The paper — 20 marks, 28 minutes

Q	TASK	MARKS	FROM
1	The faces, edges and vertices of a hexagonal prism	*3*	L175
2	The faces, edges and vertices of a square pyramid	*2*	L175
3	The surface area of a $6 \times 5 \times 4$ cm cuboid	*3*	L182–184
4	The volume of that cuboid	*2*	L176–178
5	A prism of cross-section 15 cm² and length 8 cm: its volume	*2*	L176–178
6	A cylinder of radius 4 cm and height 10 cm: its volume, $\pi = 3.14$	*3*	L176–178
7	Name the parts of a cylinder’s net and give the rectangle’s width for $r = 4$	*3*	L182–184
8	The number of cubes and the surface of a $2 \times 2 \times 2$ stack	*2*	L179–181

THE ANSWERS

8, 18, 12; *5, 8, 5*; *148* cm²; *120* cm³; *120* cm³; *502.4* cm³; two circles and a rectangle of width *25.12* cm; *8* cubes and *24* unit squares.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
hidden edges not counted	*16 edges*	*18*
the base of a pyramid forgotten	*4 faces*	*5*
surface area written in cm^3	$148\ cm^3$	$148\ cm^2$
a prism's parts added, not multiplied	$15 + 8$	$15 \cdot 8$
π left out of the cylinder	$16 \cdot 10 = 160$	$3.14 \cdot 16 \cdot 10$
the net's rectangle given width r	$4\ cm$	$2\pi r = 25.12\ cm$
every cube counted with six faces	*48*	*24*

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

IF YOU LOST MARKS ON	REVISE
Q1–Q2	counting faces, edges and vertices, L175
Q3 and Q7	nets and surface areas, L182–184
Q4–Q6	volumes of prisms and cylinders, L176–178
Q8	stacks of cubes and their views, L179–181

LOOKING FORWARD

One more Think task on nets, then the general revision of the whole Grade 6 course, and finally the presentation of the survey project.

02 Worked examples

EXAMPLE 1 Model answer, Q3: the surface area of a $6 \times 5 \times 4$ cm cuboid.

1. The three faces: 30, 20 and 24 cm².
2. $2(30 + 20 + 24)$.
3. $= 148$ cm².
   Area, so cm^2 ✓

ANSWER

148 cm²

EXAMPLE 2 Model answer, Q6: a cylinder of radius 4 cm and height 10 cm.

1. Base area $3.14 \cdot 16 = 50.24$ cm².
   Taking $\pi = 3.14$.
2. $50.24 \cdot 10$.
3. $= 502.4$ cm³.

ANSWER

502.4 cm³

EXAMPLE 3 **Model answer, Q7: the net of a cylinder of radius 4 cm.**

1. Two circles of radius 4 cm and one rectangle.
2. The rectangle is as wide as the circumference, $2\pi r$.
3. $2 \cdot 3.14 \cdot 4 = 25.12$ cm.

   Not 4 cm.

ANSWER

Two circles and a rectangle 25.12 cm wide

03 Interactive model

Return Q7 first and ask who wrote 4 cm; naming it as “the width is the circumference” settles it for the year.

The paper in eight questions

A hexagonal prism has edges:

- 12
- 16
- 18
- 24

A square pyramid has faces:

- 4
- 5
- 6
- 8

The surface area of a $6 \times 5 \times 4$ cuboid is:

- 74
- 120
- 148
- 296

Its volume is:

- 74
- 120
- 148
- 15

A prism of cross-section 15 cm² and length 8 cm holds:

- 23
- 30
- 120
- 60

A cylinder $r = 4$, $h = 10$ holds:

- 160
- 502.4
- 251.2
- 125.6

The rectangle in its net is wide:

- 4 cm
- 12.56 cm
- 25.12 cm
- 50.24 cm

A $2 \times 2 \times 2$ stack of unit cubes has surface:

- 8
- 16
- 24
- 48

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Prism	Prizma	Призма
Pyramid	Piramida	Пирамида
Cylinder	Silindr	Цилиндр
Net	Yoyilma	Развёртка
Surface area	Sirt yuzasi	Площадь поверхности
Volume	Hajm	Объём
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Q1 is easiest to get wrong on:

- the faces
- the edges
- the vertices
- the name

Q3 wants an answer in:

- cm
- cm^2
- cm^3
- litres

Q5 multiplies the cross-section by:

- three
- the length
- the surface
- six

Q6 needs:

- no π
- π once
- π twice
- a square root

Q7 asks for the rectangle's width, which is:

- the radius
- the diameter
- the circumference
- the height

Q8 counts each glued pair of faces as:

- visible
- hidden
- double
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Faces, edges and vertices of a hexagonal prism

   ANSWER $8, 18, 12$

2. Faces, edges and vertices of a square pyramid

   ANSWER $5, 8, 5$

3. The volume of a $6 \times 5 \times 4$ cm cuboid

   ANSWER 120 cm³

4. Its surface area

   ANSWER 148 cm²

5. A prism of cross-section 15 cm², length 8 cm

   ANSWER 120 cm³

6. The parts of a cylinder's net

   ANSWER Two circles and a rectangle

7. Cubes in a $2 \times 2 \times 2$ stack

   ANSWER 8

Medium · the standard question

7 PROBLEMS

1. A cylinder $r = 4$ cm, $h = 10$ cm, $\pi = 3.14$

   ANSWER 502.4 cm³

2. The width of the rectangle in that net

   ANSWER 25.12 cm

3. The surface of a $2 \times 2 \times 2$ stack of unit cubes

   ANSWER 24

4. The base area of that cylinder

   ANSWER 50.24 cm²

5. The curved surface of that cylinder

   ANSWER 251.2 cm²

6. Its total surface area

   ANSWER 351.68 cm²

7. $F + V - E$ for the hexagonal prism

   ANSWER 2

Hard · stretch and reason

7 PROBLEMS

1. Why is 16 the wrong number of edges for a hexagonal prism?

   ANSWER Two upright edges were hidden

2. Why is 160 cm³ wrong for the cylinder?

   ANSWER π was left out of the base area

3. Why is 48 wrong for the $2 \times 2 \times 2$ stack?

   ANSWER The glued faces were counted

4. A prism of volume 120 cm³ and length 8 cm: its cross-section

   ANSWER 15 cm²

5. A cuboid of surface 148 cm², 6 by 5: the third edge

   ANSWER 4 cm

6. A cylinder of volume 502.4 cm³ and radius 4 cm: its height

   ANSWER 10 cm

7. The commonest lost mark in this paper

   ANSWER The width of the cylinder's rectangle

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, with the units on every line.

1. Rewrite in full every question on which you lost a mark, naming the slip in the margin.
2. Give the faces, edges and vertices of an octagonal prism and check them.
3. Find the volume and the surface area of a cuboid 8 cm by 4 cm by 3 cm.
4. A cylinder has radius 5 cm and height 6 cm. Find its volume and its total surface area.
5. Draw a $3 \times 2 \times 1$ stack of unit cubes and give its surface area in unit squares.